CHAPTER - 1

Introduction on Mobile Application:

Mobile application development is the process to making software for smartphones and digital assistants, most commonly for Android and iOS. The software can be preinstalled on the device, downloaded from a mobile app store or accessed through a mobile web browser. The programming and markup languages used for this kind of software development include Java, Swift, C# and HTML5.

Definition:

A mobile application, also referred to as a mobile app or simply an app, is a computer program or software application designed to run on a mobile device such as a phone, tablet, or watch.

Mobile applications often stand in contrast to desktop applications which are designed to run on desktop computers, and web applications which run in mobile web browsers rather than directly on the mobile device.

A mobile application, most commonly known as an app, is a kind of application software intended to run on a mobile phone, for example, a smartphone or tablet PC.

Examples: WhatsApp, Facebook, Instagram, Tik Tok, Snapchat etc.

Mobile application Provides information and information services on the global network,including email,calendar,stock market listing and weathers information.

Key Mobile App Statistics

- Mobile apps are expected to generate over $935 billion in revenue by 2023.
- The Apple App Store has 1.96 million apps available for download.
- There are 2.87 million apps available for download on the Google Play Store.
- 21% of Millennials open an app 50+ times per day.
- 49% of people open an app 11+ times each day.

History of Mobile Application:

APP - Application Portability Profile.

- Mobile communication is so integrated into our lives that many people feel uncomfortable without a cell phone.
- Once upon a time, the most popular functions of phones were calling and sending texts.

- A smart phone is a multifunctional device that not only communicates, but helps to learn, earn, and have fun.
- This is made possible by the development of mobile applications.

✞ Mobile applications date back to the end of the **twentieth century**.
✞ Typically, they were small **arcade games, ring tone editors, calculators, calendars, and so.**

Operating systems for smart phones (Windows Mobile, Symbian, RIM, Android, Mac iOS), are open to the development of third-party software, unlike the conventional programming environment of standard cell phones.
An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs. Operating systems are found on many devices that contain a computer.

Mobile developers want the freedom to develop the powerful mobile applications users demand without restrictions. Finally, handset manufacturers want a stable, secure, and affordable platform to power their devices.



- The **Motorola DynaTAC 8000X** was the first commercially available cell phone.
- First marketed in 1983, it was 13 x 1.75 x 3.5 inches in dimension, weighed about 2.5 pounds, and allowed you to talk for a little more than half an hour.
- It retailed for $3,995, plus hefty monthly service fees and per-minute charges.
- It made calls, and there was a simple contacts application included in the operating system.

It was during this period the first "time-waster" games begin to appear.
Nokia was famous for putting the 1970s video game some of its earliest phones.
In 1997, the Nokia 6110 included a built-in version of the basic arcade game "Snake," which many consider the first mobile app.
Other followed, adding games like Pong, Tetris, and Tic-Tac-Toe.
These early phones changed the way people thought about communication. As mobile phone prices dropped, batteries improved, reception areas grew, and more and more people began carrying these handy devices.

By the late 90s, professional **Web sites** were full color and loaded with text, images, and other types of media.
They relied on **JavaScript** and **Flash** to enhance the user experience and were often designed at 800×600 pixels.

Mobile Application Distribution Platform

Mobile app distribution platforms includes **digital distribution platforms**, or marketplace '**app stores**', that are intended to provide mobile applications, aka '**apps**' to mobile devices.

App distribution is **the process of releasing an app to a broad set of users in order to promote app engagement and usage in digitally**.

The process where it is giving users to Peruse or Examine and download application, Music, Films rentals and buying and digital books created called the **digital distribution**.

Mobile Platforms:

A variety of different platforms emerged — and developers are still actively creating applications for them.
One of the first was the

- Palm OS (now Garnet OS) and RIM Blackberry OS.
- Sun Microsystems popular Java platform became Java Micro Edition (Java ME).
- Qualcomm developed its Binary Runtime Environment for Wireless (BREW). Symbian OS was developed by Nokia,
- Sony Ericsson, Motorola, and Samsung.
- The Apple iPhone iOS joined the ranks in 2007 - 2008.
- Google's Android came along a year later.
  Android and IOS are mobile platform where Google Play store and App Store are the Mobile Application Distribution Platform.

List of Mobile Application Platforms:

1. **Native platforms**
2. **Third-party platforms**

Native Platforms:

The term native app development refers to building a mobile app exclusively for a single platform. For example, you can develop a native Android app with Java or Kotlin and choose Swift and Objective-C for iOS apps. Native apps are known to deliver exceptional user experience as they are generally high performance.

Examples:

App Store Name	**Device Platform**
Amazon App Store for Android	Android and BlackBerry 10
Apple App Store for IOS/IPadOs	IOS and IPadOs
BlackBerry World	BlackBerry OS
Google Play for Android	Android
Huawei App Gallery	Huawei Android/Harmony Os
Nokia Store	Nokia Android, Symbian
Samsung Galaxy Store	Android, Tizen, Windows Mobile

Third-party platforms:

Third-party platforms are software distribution platforms which are used as alternatives for operating system native distribution platforms.

Independent operating systems are software collections which use their own software distribution, customised user interface (UI), software development kit (SDK) and application programming interface (API) (except billing API which is related only to application store).

Examples:

App Store Name	**Device Platform**
Appland	Android and iOS
Aptoide	Android
Cydia	IOS
Cafe Bazaar	Android
GetJar	Android, BlackBerryOS, Flash Lite, Java, iOS, Palm OS, Symbian, Windows Phone
Opera Mobile Store	Android, Java, Symbian, BlackBerry OS, Windows Mobile, iOS

Types of Mobile Application:

1. Native app
2. Web-based app
3. Hybrid app

- **Native apps are created for one specific platform or operating system.**

  Native apps are built specifically for a mobile device's operating system (OS). Thus, you can have native Android mobile apps or native iOS apps, not to mention all the other platforms and devices. Because they're built for just one platform, you cannot mix and match – say, use a Blackberry app on an Android phone or use an iOS app on a Windows phone.



- **Web apps are responsive versions of websites that can work on any mobile device or OS because they're delivered using a mobile browser.**

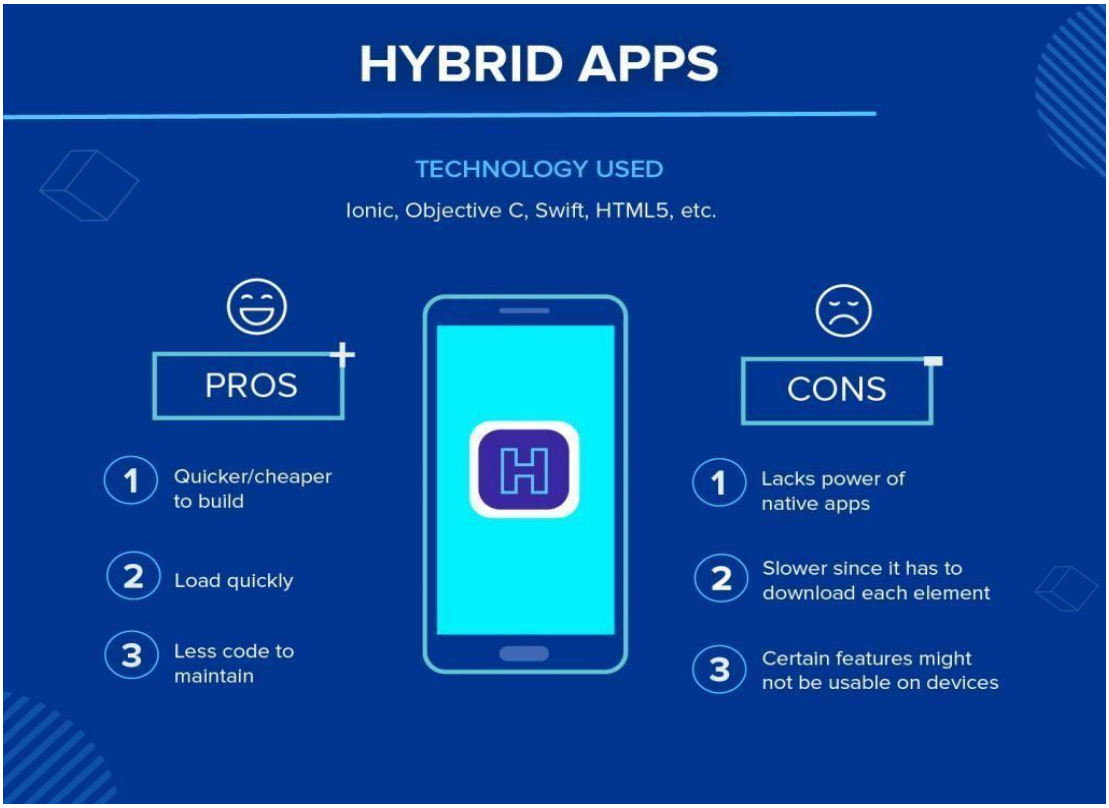
  Web apps behave similarly to native apps but are accessed via a web browser on your mobile device. They're not standalone apps in the sense of having to download and install code into your device. They're actually responsive websites that adapt its user interface to the device the user is on. In fact, when you come

across the option to "install" a web app, it often simply bookmarks the website URL on your device.



- **Hybrid apps are combinations of both native and web apps, but wrapped within a native app, giving it the ability to have its own icon or be downloaded from an app store.**

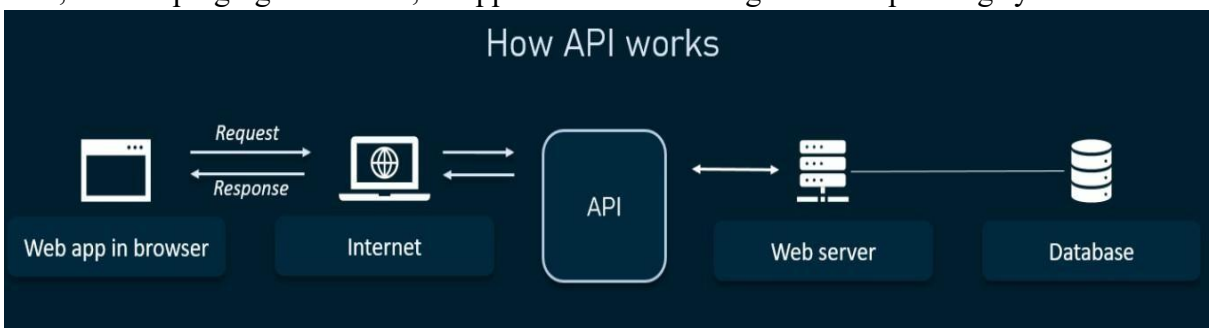
  And then there are the hybrid apps. These are web apps that look and feel like native apps. They might have a home screen app icon, responsive design, fast performance, even be able to function offline, but they're really web apps made to look native.



Examples:
Marketwatch (built with Ionic)
Sworkit (built with PhoneGap)
Untappd (built with PhoneGap)
Giro D'Italia official app (built with Onsen UI)
Offcourse Golf (built with Onsen UI)
Tripline (built with Framework 7)

What is an API ?

API stands for "application programming interface."
An API is essentially a set of rules that dictate how two machines talk to each other. Some examples of API-based interactions include a cloud application communicating with a server, servers pinging each other, or applications interacting with an operating system.



Google Play:

Play Store is Google's official pre-installed app store on Android-certified devices. It provides access to content on the Google Play Store, including apps, books, magazines, music, movies, and television programs.

Google Play Store and formerly Android Market, is a digital distribution service operated and developed by Google. It serves as the official app store for certified devices running on the Android operating system and its derivatives as well as Chrome OS, allowing users to browse and download applications developed with the Android software development kit (SDK) and published through Google. Google Play also serves as a digital media store, offering music, books, movies, and television programs. Content that has been purchased on Google Play Movies & TV and Google Play Books can be accessed on a web browser, and through the Android and iOS apps. Applications are available through Google Play either free of charge or at a cost.

[**Source Wikipedia**].

Platform : Android, Android TV, Wear OS, Chrome OS, Web;



Apple Store:

Apple Inc. is an American multinational technology company that specializes in consumer electronics, computer software and online services.

The Apple App Store is a digital distribution platform where individuals can buy and download digital software and applications. Apps—an abbreviation of the word "applications"—are software tools that provide additional functionality to an operating system. Apps purchased from the Apple App Store are stored in the iCloud—a cloud storage and cloud computing service from Apple—for easy access from any signed-in device.

While Apple maintains that it holds a trademark on the term “app store," the term is used to reference any platform where apps are sold. Examples of other app stores are Google's Google Play, the Amazon Appstore, Blackberry World, and Microsoft's Windows Store.

The Apple App Store opened on July 10, 2008, the day before the launch of the iPhone 3G. Apple’s first app store was originally intended for iOS, but it was later expanded to provide apps for Macs with the App Store macOS in early 2011.



Windows Store:

Windows Store is a digital distribution platform owned by Microsoft. It started as an app store for Windows 8 and Windows Server 2012 as the primary means of distributing Universal Windows Platform apps. With Windows 10, Microsoft merged its other distribution platforms (Windows Marketplace, Windows Phone Store, Xbox Music, Xbox Video, Xbox Store, and a web storefront also known as "Microsoft Store") into Microsoft Store, making it a unified distribution point for apps, console games, and digital videos.

The Microsoft Store is an app on Windows and allows you to install apps and games, purchase digital media and Microsoft devices.

Microsoft announced the Windows Store in September 2011 and opened the marketplace in February 2012. As of October 2012, the Windows Store contained about 3,600 apps.

The Windows Store does not sell apps for the Windows Phone platform. Those apps are available through the Windows Phone Store.



Google Assistant

Google Assistant is an artificial intelligence–powered virtual assistant developed by Google that is primarily available on mobile and smart home devices. Unlike the company's previous virtual assistant, Google Now, the Google Assistant can engage in two-way conversations.

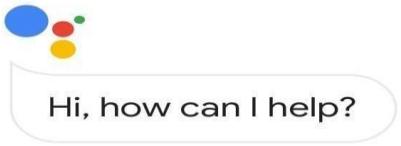
Google Assistant offers voice commands, voice searching, and voice-activated device control, letting you complete a number of tasks after you've said the "OK Google" or "Hey Google" wake words. It is designed to give you conversational interactions.

Google Assistant will:

- Control your devices and your smart home
- Access information from your calendars and other personal information
- Find information online, from restaurant bookings to directions, weather and news
- Control your music
- Play content on your Chromecast or other compatible devices
- Run timers and reminders
- Make appointments and send messages
- Open apps on your phone
- Read your notifications to you
- Real-time spoken translations
- Play games

Google Assistant originally launched on the Google Pixel smartphones and Google Home, but it is now available to just about all modern Android devices, including Wear OS devices,

Android TV, and Nvidia Shield, as well as any cars that support Android Auto and other devices too, like Nest cameras and the Lenovo Smart Clock



Siri:

As we mentioned, Siri is a built-in, voice-controlled personal assistant available for Apple users. The idea is that you talk to her as you would a friend and she aims to help you get things done, whether that be making a dinner reservation or sending a message.

Siri is designed to offer you a seamless way of interacting with your iPhone, iPad, iPod Touch, Apple Watch, HomePod or Mac by you speaking to her and her speaking back to you to find or do what you need. You can ask her questions, tell her to show you something or issue her with commands for her to execute on your behalf, hands-free.

Siri is based on the based on the fields of Artificial Intelligence and Natural Language Processing, and it is comprised of three components - a conversational interface, personal context awareness and service delegation.



<u>Difference between Software and Application:</u> [Source GFG] Software:

Software is a collection of programs that co-ordinates with the hardware to run the machine. It is set of instructions or data that operates the computer how to work. Software is opposite of the physical aspect, i.e., hardware and it is a complement to the hardware in computer.

Application:

Application is package that performs a specific task for end users. It is a product or a program that is designed only for end users requirements. All the applications may be in category of software but vice-versa is not possible.

Software	Application

Software is a set of instructions or data that operates the hardware.	Application is a package to perform a specific task.
Software is an all encompassing term for computer data.	Application is a type of software that does a certain task.
Software may executable or may not be.	Application is always executable.
Software is not often operating system based.	Application is operation system based.
Software does not need user interaction for functioning.	Application need user interaction for functioning.
Software is used as mediator between user and hardware.	Application is used only by end users.
All the software are not applications.	All the application are software.

Difference Between Mobile Apps and Web Apps:

Mobile Apps	**Web Apps**
Mobile apps are developed for a specific platform, such as iOS for the Apple iPhone or Android	Web Apps are accessed via the internet browser and will function according to the device you're viewing them.
They are downloaded and installed via an app store such as Google Play Store and Apple Store and have access to system resources, such as GPS and the camera of the device.	Web apps are not native to a particular system and there is no need to be downloaded or installed.
Mobile apps may work offline.	In order to run web apps need an active internet connection.
Mobile Apps are comparatively faster.	Web Apps are comparatively slower.
Mobile apps have more safety and security.	Web apps have comparatively low security.
Maintaining and consistently update of Mobile apps cause more cost.	In Web Apps App store approval is not required, so web apps can be launched easily.

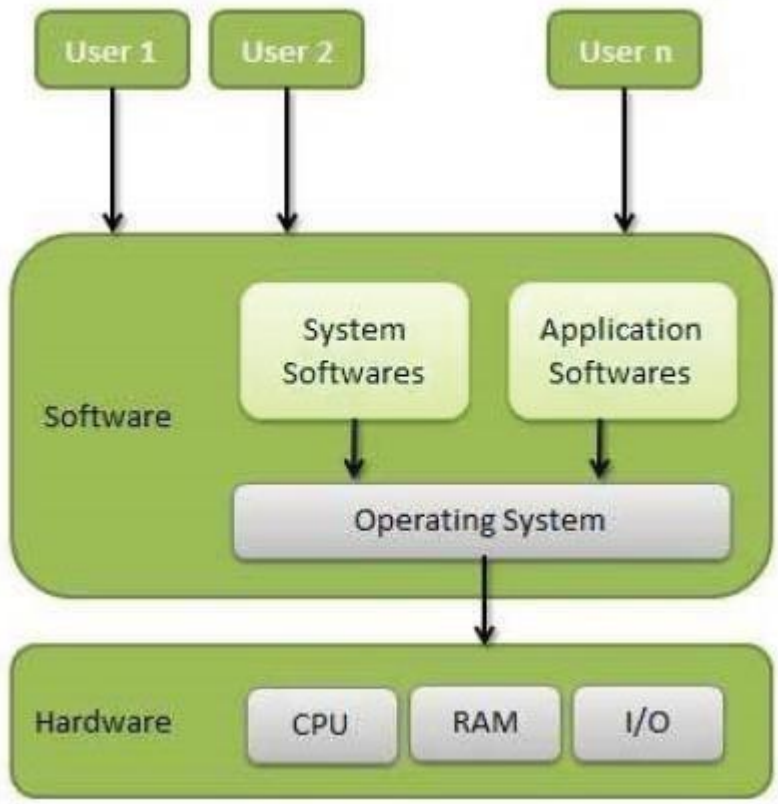
CHAPTER - 2

Mobile Operating Systems:

Before Mobile Operating System understood about Operating system:

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory controlling memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

Some OperatingSome popular Operating Systems include Linux Operating System, Windows OperatingSystem, System, VMS, OS/400, AIX, z/OS, etc.



Operating system goals:

- Make the computer system convenient to use. It hides the difficulty in managing the hardware.
- Use the computer hardware in an efficient manner
- Provide and environment in which user can easily interface with computer.
- It is a resource allocator.

Mobile Operating System:

A mobile operating system is an operating system that helps to run other application software on mobile devices. It is the same kind of software as the famous computer operating systems like Linux and Windows, but now they are light and simple to some extent.

The operating systems found on smartphones include Symbian OS, iPhone OS, RIM's BlackBerry, Windows Mobile, Palm WebOS, Android, and Maemo. Android, WebOS, and Maemo are all derived from Linux. The iPhone OS originated from BSD and NeXTSTEP, which are related to Unix.

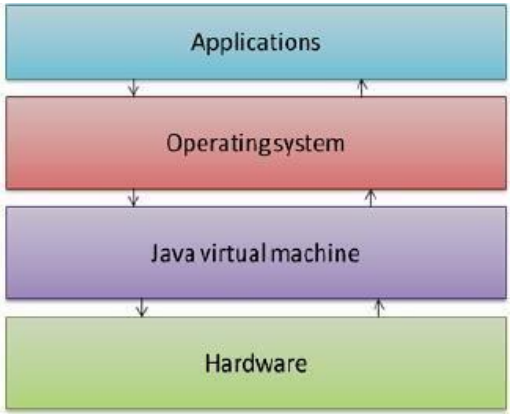


Fig.: Blackberry Mobile Operating system

- Operating system :✓ Controls and coordinates use of hardware among various applications and users
- Application programs – define the ways in which the system resources are used to solve the computing problems of the users
- Users : ✓ People, machines, other computers
- JVM (Java Virtual Machine) is an abstract machine. It is a specification that provides runtime environment in which java bytecode can be executed.

JVMs are available for many hardware and software platforms (i.e. JVM is platform dependent).

Hardware – provides basic computing resources

✓ CPU, memory, I/O devices

Mobile virtualization:

Mobile virtualization is hardware virtualization on a mobile phone or connected wireless device. It enables multiple operating systems or virtual machines to run simultaneously on a mobile phone or connected wireless device.

Kernel in Mobile Operating System

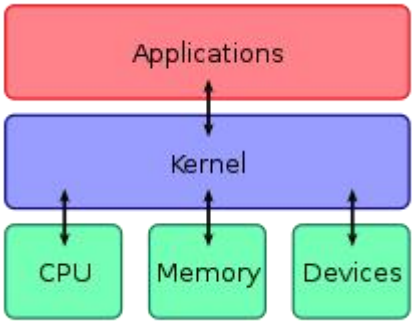
Kernel is central component of an operating system that manages operations of computer and hardware. It basically manages operations of memory and CPU time. It is core component of an operating system. Kernel acts as a bridge between applications and data processing performed at hardware level using inter-process communication and system calls.

Kernel loads first into memory when an operating system is loaded and remains into memory until operating system is shut down again. It is responsible for various tasks such as disk management, task management, and memory management.

It decides which process should be allocated to processor to execute and which process should be kept in main memory to execute. It basically acts as an interface between user applications and hardware. The major aim of kernel is to manage communication between software i.e. user-level applications and hardware i.e., CPU and disk memory.

Objectives of Kernel:

1. To establish communication between user level application and hardware.
2. To decide state of incoming processes.
3. To control disk management.
4. To control memory management.
5. To control task management.



Needs of Mobile Operating System:

Goal of an Mobile Operating System:

The fundamental goal of a Mobile System is to execute user programs and to make tasks easier. Various application programs along with hardware systems are used to perform this work. Operating System is software that manages and controls the entire set of resources and effectively utilizes every part of a Mobile system. **Need of Mobile Operating System:**

1. **OS as a platform for Application programs:**

Operating system provides a platform, on top of which, other programs, called application programs can run. These application programs help the users to perform a specific task easily. It acts as an interface. It is designed in such a manner that it operates, controls, and executes various applications.

2. **Managing Input-Output unit:**

Operating System also allows the system to manage its own resources such as memory, display screen or a speaker, a data storage device,keyboard,microphone etc. Management of these resources is required for effective utilization. The operating system controls the various system input-output resources and allocates them to the users or programs as per their requirement.

3. **Consistent user interface:**

Operating System provides the user an easy-to-work user interface, so the user doesn't have to learn a different UI every time and can focus on the content and be productive as quickly as possible. Operating System provides templates, UI components to make the working of a devices, really easy for the user.

4. **Multitasking:**

Operating System manages memory and allows multiple programs to run in their own space and even communicate with each other through shared memory. Multitasking gives users a good experience as they can perform several tasks on a devices at a time.

Features of Mobile Operating System

1. Easy to use

The graphics should be attractive.
The buttons and features should be easy to use. mOreover, the functionalities should not be very compilicated.
Features should be powerful and useful.

2. Good app store

An app is one of the basic part of an OS.
Good and useful apps forms an important part of an OS.
The apps should be simple and interactive.

3. Good battery life

Power is one of the main requirement of a smartphone.
They require power for processors sensors etc. Therefore, the battery holds a very important role.

Smartphones power usage keeps on increasing therefore, a good battery backup is very essential.

4. Data usage and organization

An operating system should focus on controlling the data and network usage. It should keep the limit and requirement in focus.

Secondly, the organization of data related to to-do lists, calendars, alarms, reminders etc is very important. A good OS should keep this data in a very organized and safe manner. Moreover, the data should be readily and easily available.

Types of Mobile Operating system:

1. Android OS:

The Android operating system is the most popular operating system today. It is a mobile OS based on the Linux Kernel and open-source software. The android operating system was developed by Google. The first Android device was launched in 2008.

Android is a mobile operating system based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets. Android is developed by a consortium of developers known as the Open Handset Alliance and commercially sponsored by Google. It was unveiled in November 2007, with the first commercial Android device, being launched in September 2008.

It is free and open-source software; its source code is known as Android Open Source Project (AOSP), which is primarily licensed under the Apache License.



2. IOS:

The iOS was developed by the Apple inc for the use on its device. The iOS operating system is the most popular operating system today. It is a very secure operating system. The iOS operating system is not available for any other mobiles.

iOS (formerly iPhone OS) is a mobile operating system created and developed by Apple Inc. exclusively for its hardware. It is the operating system that powers many of the company's mobile devices, including the iPhone and iPod Touch; the term also included the versions running on iPads until the name iPadOS was introduced with version 13 in 2019. It is the world's second-most widely installed mobile operating system, after Android. It is the basis for three other operating systems made by Apple.

Unveiled in 2007 for the first-generation iPhone, iOS has since been extended to support other Apple devices such as the iPod Touch (September 2007) and the iPad (introduced: January 2010; availability: April 2010.) As of March 2018, Apple's App Store contains more than 2.1 million iOS applications, 1 million of which are native for iPads.



3. BlackBerry OS:

The BlackBerry operating system is a mobile operating system developed by Research In Motion (RIM). This operating system was designed specifically for BlackBerry handheld devices. This operating system is beneficial for the corporate users because it provides synchronization with Microsoft Exchange, Novell GroupWise email, Lotus Domino, and other business software when used with the BlackBerry Enterprise Server.

BlackBerry OS is a proprietary mobile operating system developed by Canadian company BlackBerry Limited for its BlackBerry line of smartphone handheld devices. The operating system provides multitasking and supports specialized input devices that have been adopted by BlackBerry for use in its handhelds, particularly the trackwheel, trackball, and most recently, the trackpad and touchscreen. Initial release January 19, 1999;



CHAPTER - 3

Android Mobile Operating System

INTRODUCTION TO ANDROID

Android is a mobile operating system based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets. Android is developed by a consortium or an association of developers known as the Open Handset Alliance and commercially sponsored by Google. It was unveiled or show or announce publicly for the first time. in November 2007, with the first commercial Android device, the HTC Dream, being launched in September 2008.

In 2007, Apple launched the first iPhone and ushered or **begin** in a new era in mobile computing. In Sept. 2008, the very first Android smartphone was announced, the T-Mobile G1 went on sale in the U.S. Oct. of that year.

Android 1.0 OS inside integrated a number of the company's other products and services, including Google Maps,

YouTube, and an HTML browser (pre-Chrome) that, of course, used Google's search services. It also had the first version of Android Market, the app store with ―dozens of unique, firstofa-kind Android applications.

The first version of the OS (1.0) released in Sept. 2008 did not have a code name at all. However, it reportedly used the internal name ―Petit four while it was in development at Google. The name refers to a French dessert.

Android has come a long way from its humble beginnings, as the product of a small startup, all the way to becoming the leading mobile operating system worldwide. Google's introduction of Project Treble in Android Oreo should make it easier for phone makers to update their devices faster.

One challenge for Android device owners that has been an issue for the OS ever since it launched is updating it with the latest security patches, for major feature updates. Google's supported Nexus and Pixel devices consistently receive regular monthly security updates, and the latest version of the OS.

HISTORY OFANDROID

In October 2003, Android Inc was founded in Palo Alto, California and its four founders were Rich Miner, Nick Sears, Chris White, and Andy Rubin. In 2005, Android was acquired by Google. Rubin stayed at Google as head of the Android team until 2013.

The logo for the Android OS was created by Irina Blok while she was employed by Google.

Versions of Android along with their names

ANDROID 1.0:	ALPHA
ANDROID 1.1:	BETA
ANDROID1.5	CUPCAKE
ANDROID1.6	DONUT
ANDROID2.0-2.1	ÉCLAIR
ANDROID2.2	FROYO
ANDROID2.3	GINGERBREAD
ANDROID3.0	HONEYCOMB

ANDROID4.0	ICECREAMSANDWICH
ANDROID4.1-4.3	JELLYBEAN
ANDROID4.4	KITKAT
ANDROID5.0	LOLLIPOP
ANDROID6.0	MARSHMALLOW
ANDROID7.0	NOUGAT
ANDROID8.0	OREO
ANDROID9.0	PIE
ANDROID10	Q

ANDROID 11

Android Q will allow users to control apps' access to their phone's Photos and Videos or the Audio collections via new runtime permissions.

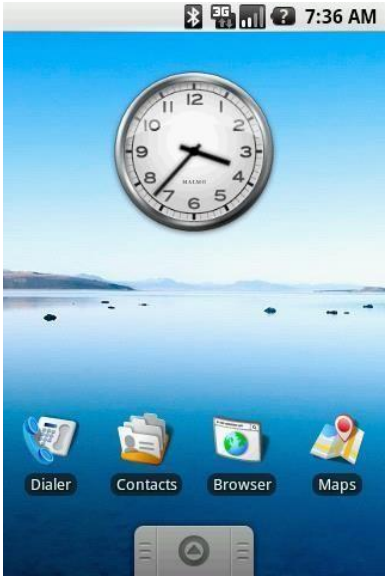
Description of version of mobile operating system:

1. Android 1.0: Alpha

The initial version of Android was released on September 23, 2008, known as Android 1.0. There were many loopholes in this alpha version including the necessity of physical keyboards or hardware buttons. It has an android API (application program interface) level 1. API level is the integer value that identifies the API framework revision provided by the version of the android platform. API framework includes the core set of classes, packages, XML elements, intents, and permissions that the application can request.

Features

- Google Maps, browser, calendar
- Camera and scroll down the notification bar
- Gmail integration, Contacts, and Google Synchronization.
- Wireless supports – Wi-Fi and Bluetooth.



2. Android 1.1: Beta

Android Beta version was released on February 9, 2009, with API changes and resolved issues encountered in 1.0. This android version was released for T-Mobile G1 devices only.

Features

- Display details and reviews for locations
- Add Save attachment in the message
- Provide detailed information by clicking on the business

3. Android 1.5: Cupcake

It was released on April 30, 2009, with the first official public code name and amazing features as compared to the old versions. Moreover, it also brought the third-party app widgets that was the most distinguishing and valuable feature.

Features

- on-screen keyboard and search function
- Uploading videos and images
- Copy and paste facility and video recordings
- Support for MPEG4 and 3GP formats.



4. Android 1.6: Donut

The android 1.6 version was released on September 15, 2009, was many valuable changes including the ability to operate on a number of different screen resolutions and sizes. Donut provides the voice and text entry search including bookmark history, and the ability to select multiple images for deletion.

Features

- Power Control widget for handling Wi-Fi, Bluetooth, GPS, etc.
- Gallery and Camera quick toggling features
- WVGA screen resolution speed
- Technology support for CDMA/EVDO, 802.1x, VPNs, and a text-to-speech engine
- Speed improvements for camera and searching applications
- Quick Search Box



5. Android 2.0: Eclair

Éclair 2.0 – 2.1 was released on December 3, 2009, with multiple account synchronizations of email and contacts. Moreover, there are many new features were added including flash support, scene mode, white balance, color effect, macro focus and digital zoom. Android 2.1 version brought bug fixes and stability improvements to the Éclair.

Features

- Update UI
- Support Bluetooth 2.1
- Improve Google map
- Minor API Changes
- Support Live and animated Wallpapers
- Ability to add contact's photo and select to call, message or email.



6. Android 2.2: Froyo

Android version 2.2 was released on May 20, 2010, with speed, memory, and performance improvements. It was introduced with the enhanced Bluetooth functionality and compatibility with docks, portable WiFi hotspot for sharing the 3G connection. Google Nexus One was the first smartphone having an android 2.2 android version. Android version list provides you the complete historical knowledge of android operating system updates and improvements.

Features

- Support for Animated GIF and multiple keyboard languages
- Speed and performance improvements
- Upload file support in the browser
- Support numeric & alphanumeric passwords to enhance security
- Increased Compatibility with car kits and headsets
- Wi-Fi Support Hotspot functionality



7. Android 2.3: Gingerbread

Gingerbread exists in the 7th position in the android version list that was released on December 6, 2010. The main enhanced feature was the introduction of gaming API with improved graphical intense gaming, so it has boomed the mobile games.

Features

- Improve Copy and Paste Facility
- Updated UI design
- Support for VP8 and WebM video format

- Video calling and Social Networking Supports
- Easy to use a keyboard with faster and intuitive typing



8. Android 3.0: Honeycomb

Honeycomb from the android version list was released in 2011 that was tablet-only release to launch the Motorola Xoom. It was also suitable for those mobiles having a larger view than current smartphones.

Features

- Gmail, contacts, camera and gallery improvements
- Support for passwords with complex characters
- encrypted storage and updated 3D UI
- Supports multiprocessors and recent apps for easy visual multitasking
- Media Sync from SD Card
- Action bar for application control
- System bar for global status and notifications
- Google eBooks and Talk Video Chat
- Support Adobe Flash in Browser
- More sensor support
- High-performance Wi-Fi Connections and Lock • Chinese handwriting and redesigned keyboard



9. Android 4.0: Ice Cream Sandwich

The ice cream sandwich was released on October 19, 2011, with many enhanced features to enter the era of modern design. The snapshot was introduced to take screenshots by holding the volume and power button. Android version list has a variety of android operating systems but as compared to all older versions, ice cream sandwich widgets are more robust and resizable.

Features

- Spelling check feature
- Wi-Fi direct
- Photo Decor facility and on-screen buttons
- Unlocking with face-fixing.
- Card-like appearance for app-switching
- Improved video recording with high resolution
- Better Camera performance
- Ability to open up to 16 tabs in the web browser



10. Android 4.1: Jelly Bean

The better version of android known as Jelly Bean was released in June 2012 with Google Digital Assistant technology accessible from the home screen. The spectacular predictive intelligence utility provides expandable and interactive notifications. Moreover, users can enjoy the multi-user support into play and many other valuable features.

Features

- Voice search and typing
- Panorama
- Project Butter
- Expandable notifications
- Daydream as a screensaver
- Power control
- Support USB audio
- Improved camera app
- Security improvements
- New gestures and accessibility features
- Multiple user accounts (Only for tablets)

- 4k resolution support
- Supporting Bluetooth with low energy
- Bi-directional text and different language support
- Set or adjust the volume of incoming calls and show a message alert
- Google displays relevant content based on your search history • Native emoji support



11. Android 4.4: KitKat

Android version 4.4 was released on September 3, 2013, with more focused on better user experience. KitKat is optimized to run at a larger range of old versions from the android version list. The smartphone must have a minimum of 512 MB of RAM.

Features

- Screen Recording
- Contact Prioritization
- GPS Support
- Smarter Caller ID
- Offline music support
- UI updates for alarm and google map navigations
- Cartoonish ideograms and emojis to the Google keyboard
- KitKat has 'OK Google' feature that allows access to Google to the users without touching your smartphones.



12. Android 5.0: Lollipop

Lollipop or android version 5.0 was released on November 12, 2014, with a redesigned user interface and built with "material design". It gives a new and modern look extended across all of android, applications, and google products. Lollipop from the android version list comes with many amazing features including Support for better notification management.

Features

- Support ART
- Better device protection
- Notifications can be flicked away from the lock screen
- Better and improved UI
- Built-in battery saver feature
- New material design
- Revamped navigation bar • Support for Multiple sim cards
- High definition of voice call.



13. Android 6.0: Marshmallow

In the year 2015, Google used "Macadamia Nut Cookie" to describe android version 6.0 before the Marshmallow official announcement.

Features

- Support for Fingerprint readers
- Type C USB support
- Multi-window experience
- 'Sleep Mode' for saving battery life
- Clear permission system
- Custom google tabs and improved Copy-pasting



14. Android 7.0: Nougat

The android 7.0 was released in 2016 with a native split-screen mode, data saver functionality, and a "bundled-by-app" system to organize notifications.

Features

- Provide multitasking and split-screen mode
- Storage manager enhancements
- Quick setting toggles
- Display touch enhancements
- Better setting application
- Inline reply to messages and notifications without opening applications



15. Android 8.0: Oreo

Oreo was released in the year 2017 having native picture-in-picture mode, notification snoozing options and better control over how applications can alert you by notifications.

Features

- Password autofill
- Auto-enable Wi-Fi
- Downloadable fonts
- Multi-display support
- Support Picture-in-Picture
- Notification channels and snooze notification
- Google Play support and new emoji styling
- Adaptive icons and smart text selection



16. Android 9: Pie

The pie was publicly released on August 6, 2018, with plenty of amazing features according to the users' interests and requirements. According to a report, Pie is the most widely used operating system in the android version list.

Features

- Sound amplifier with select to speak options
- Artificial intelligence (AI) compatibility
- Adaptive Battery and Brightness with background restrictions
- Multi-camera support with the external camera compatibility
- New Gesture Navigation and App Actions
- New Screenshot Shortcut key and accessibility menu
- Easier Screen Rotation and edge to edge screens support
- Volume and Sound enhancements
- Selectable Dark Mode
- HDR, HD audio, multiple Bluetooth connections
- Slices and long press to overview selection
- Improved Security features for extra protection
- Digital Wellbeing with app timers, dashboard and do not disturb options
- Android backups and privacy enhancements
- More Notification Information and easier text selection



17. Android 10: Android Q

Android version 10 is officially released on September 3, 2019, with enhanced features and functionalities with higher API levels.

Features

- Support for foldable smartphones with flexible displays
- Dark mode for eyes comfortability
- Navigation control over gesture quicker and intuitive ever
- Sound amplifier with more clear sound
- Smart reply suggestions for all messaging apps
- Live caption for media playing on a smartphone
- Undo app removal
- Better notification control with many options

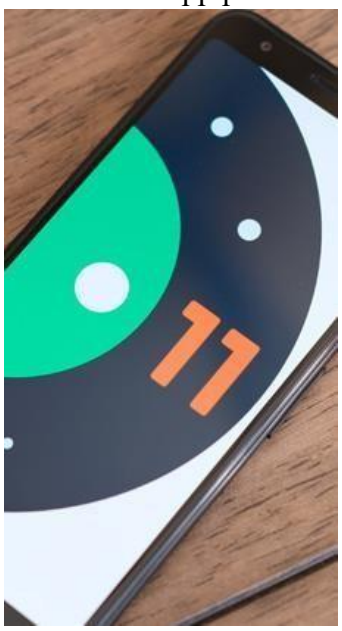


18. Android 11: (API 30)

Android developers are continually working to provide more advanced applications as per the user requirements. Most of the developers are searching Android Developer vs Web Developer to choose a trending field.

The following are the new features that you will experience in the new android 11 that is going to release and will be the latest version in the android version list.

- Native screen recording
- Muting notifications during video
- Increase touch sensitivity
- Notification History
- Auto revoke app permissions



ANDROID BRIEF:

Android programming is based on Java programming language so if you have basic understanding on Java programming then it will be a fun to learn Android application development.

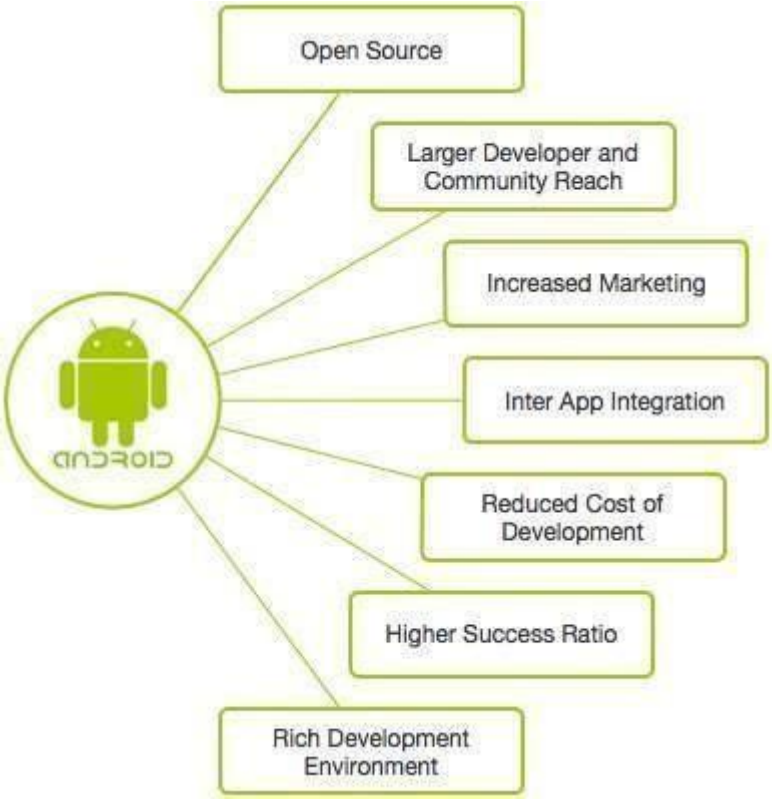
Android is an open source and Linux-based Operating System for mobile devices such as smartphones and tablet computers. Android was developed by the Open Handset Alliance, led by Google, and other companies.

Android offers a unified approach to application development for mobile devices which means developers need only develop for Android, and their applications should be able to run on different devices powered by Android.

The first beta version of the Android Software Development Kit (SDK) was released by Google in 2007 where as the first commercial version, Android 1.0, was released in September 2008.

On June 27, 2012, at the Google I/O conference, Google announced the next Android version, 4.1 Jelly Bean. Jelly Bean is an incremental update, with the primary aim of improving the user interface, both in terms of functionality and performance. The source code for Android is available under free and open source software licenses. Google publishes most of the code under the Apache License version 2.0 and the rest, Linux kernel changes, under the GNU General Public License version 2.

Why prefer Android?



Feature

1. Beautiful UI
   Android OS basic screen provides a beautiful and intuitive user interface.

2. Connectivity
   GSM/EDGE, IDEN, CDMA, EV-DO, UMTS, Bluetooth, Wi-Fi, LTE, NFC and WiMAX.

3. Storage
   SQLite, a lightweight relational database, is used for data storage purposes.

4. Media support
   H.263, H.264, MPEG-4 SP, AMR, AMR-WB, AAC, HE-AAC, AAC 5.1, MP3, MIDI, Ogg Vorbis, WAV, JPEG, PNG, GIF, and BMP.

5. Messaging
   SMS and MMS

6. Web browser
   Based on the open-source WebKit layout engine, coupled with Chrome's V8 JavaScript engine supporting HTML5 and CSS3.

7. Multi-touch
   Android has native support for multi-touch which was initially made available in handsets such as the HTC Hero.

8. Multi-tasking
   User can jump from one task to another and same time various application can run simultaneously.

   ETC.

The code names of android ranges from **A to N** currently, such as Aestro, Blender, Cupcake, Donut, Eclair, Froyo, Gingerbread, Honeycomb, Ice Cream Sandwitch, Jelly Bean, KitKat, Lollipop and Marshmallow.

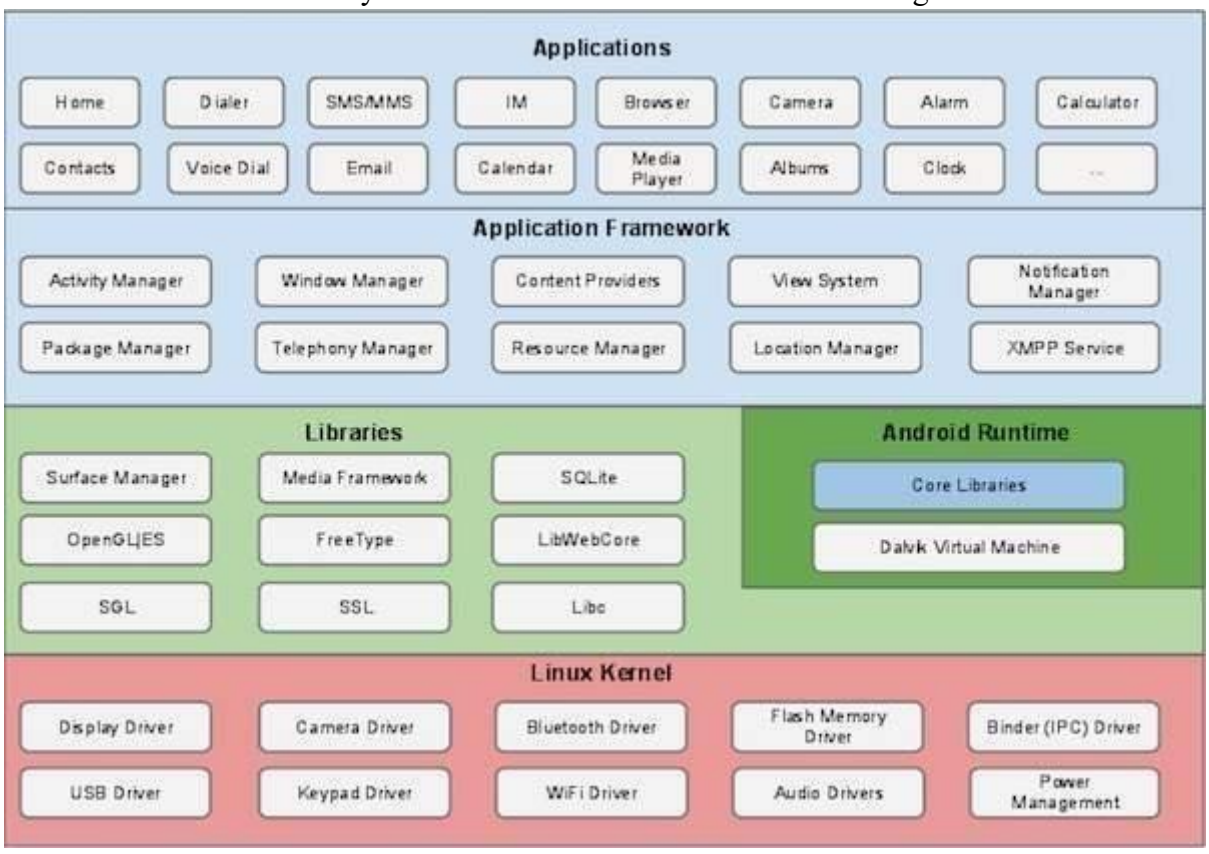
Android application development on either of the following operating systems −

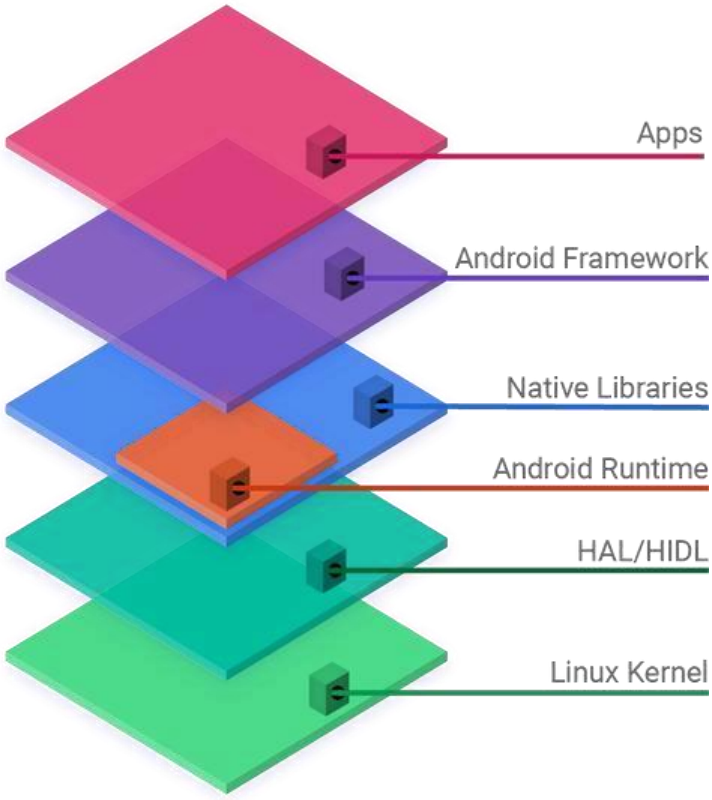
1. Microsoft Windows XP or later version.

2. Mac OS X 10.5.8 or later version with Intel chip.

3. Linux including GNU C Library 2.7 or later.

Android operating system is a stack of software components which is roughly divided into five sections and four main layers as shown below in the architecture diagram.





Interface Definition Language

Linux kernel

At the bottom of the layers is Linux - Linux 3.6 with approximately 115 patches. This provides a level of abstraction between the device hardware and it contains all the essential hardware drivers like camera, keypad, display etc. Also, the kernel handles all the things that Linux is really good at such as networking and a vast array of device drivers, which take the pain out of interfacing to peripheral hardware.

Libraries

On top of Linux kernel there is a set of libraries including open-source Web browser engine WebKit, well known library libc, SQLite database which is a useful repository for storage and sharing of application data, libraries to play and record audio and video, SSL libraries responsible for Internet security etc.

Android Libraries

This category encompasses those Java-based libraries that are specific to Android development. Examples of libraries in this category include the application framework libraries in addition to those that facilitate user interface building, graphics drawing and database access. A summary of some key core Android libraries available to the Android developer is as follows − android.app − Provides access to the application model and is the cornerstone of all Android applications.

android.content − Facilitates content access, publishing and messaging between applications and application components.

android.database − Used to access data published by content providers and includes SQLite database management classes.

android.opengl − A Java interface to the OpenGL ES 3D graphics rendering API.

android.os − Provides applications with access to standard operating system services including messages, system services and inter-process communication.

android.text − Used to render and manipulate text on a device display.

android.view − The fundamental building blocks of application user interfaces.

android.widget − A rich collection of pre-built user interface components such as buttons, labels, list views, layout managers, radio buttons etc.

android.webkit − A set of classes intended to allow web-browsing capabilities to be built into applications.

Having covered the Java-based core libraries in the Android runtime, it is now time to turn our attention to the C/C++ based libraries contained in this layer of the Android software stack.

Android Runtime

This is the third section of the architecture and available on the second layer from the bottom. This section provides a key component called Dalvik Virtual Machine which is a kind of Java Virtual Machine specially designed and optimized for Android.

The Dalvik VM makes use of Linux core features like memory management and multithreading, which is intrinsic in the Java language. The Dalvik VM enables every Android application to run in its own process, with its own instance of the Dalvik virtual machine.

The Android runtime also provides a set of core libraries which enable Android application developers to write Android applications using standard Java programming language.

Application Framework

The Application Framework layer provides many higher-level services to applications in the form of Java classes. Application developers are allowed to make use of these services in their applications.

The Android framework includes the following key services −

Activity Manager − Controls all aspects of the application lifecycle and activity stack.

Content Providers − Allows applications to publish and share data with other applications.

Resource Manager − Provides access to non-code embedded resources such as strings, color settings and user interface layouts.

Notifications Manager − Allows applications to display alerts and notifications to the user.

View System − An extensible set of views used to create application user interfaces.

Applications
You will find all the Android application at the top layer. You will write your application to be installed on this layer only. Examples of such applications are Contacts Books, Browser, Games etc.

ADVANTAGES OF ANDROID OPERATING SYSTEM

Open Ecosystem

Android has a lot of app stores and one of them is in the shape of Google Play (the official app store provided by Google), which comes as standard on almost every smartphone and tablet in order for users to find and download new software.

Customizable UI

Google's been working hard to make Android's user interface (UI) as flexible and customizable as possible. In doing so Google's loaded Android with a host of customizable widgets. These can be used anywhere in UI and they offer updates or shortcuts to a variety of services, such as email, calendar, and massages etc.

Open Source

The Android operating system allows the developers and hardware manufacturers to make changes to the operating system's core software. This allows the companies to make changes in the OS to work in very specific industries.

Innovations Reach the Market Quicker

Android platform has a proven track record of supporting the latest cutting-edge ideas. While casual apps still seem to appear on almost every operating system at the same time, the bigger hardware innovations almost start at Google.

Customized Roms

There are a lot of third-party apps that give you advanced features on Android, one of the best things is that people can use it, modify it all over, and install custom versions instead of the

one that comes with your phone. It will allow you a lot of system-level tweaks that you would not be able to get this easily on any other OS like Windows or IOS.

Affordable Development

You can easily get your own app out in the android market and you don't need to pay big development fees. The application environments are free of charge and you can save a lot of money and there is no limit which means that if you want to make more than one app, you can do that without any problems.

APP Distribution

With iOS apps, you need to use the Apple App Store. With Android apps, you don't need any of that. Just publish the app where you wish. There's no "empire" looking over your shoulder. There are a lot of third-party stores and you can publish your app very easily.

Affordable

Since Android OS is affordable to buy so most of the customers love to buy android phones because of the low price of android phones. iOS and windows phones are more expensive as compared to android phones.

DISADVANTAGES OF ANDROID OPERATING SYSTEMS

Less memory storage

Android supports external storage memory but due to the small internal system storage it is very difficult for an android user to download heavy applications or work online with game like applications.

Automatic closing of Apps/Games

The main drawback of Android Operating system is that large apps are forced by OS to close and it is very annoying for the android user.

Data connection

Android apps are processing most of the time in the background and leads to data connection problems.

Battery life

Android apps are processing most of the time in the background and leads to the battery problems.

INTRODUCTION TO ANDROID DEVELOPMENT TOOLKIT

Android is a development tool that lets you perform these tasks: Manage Android Virtual Devices (AVD) Create **and update** Android projects. Update your sdk with new platform addons and documentation.

Android App Development Tools & Software

1. Android Studio.
2. Android Debug Bridge (ADB)
3. Android Virtual Device (AVD) Manager.
4. Eclipse.
5. Fabric.
6. FlowUp.
7. GameMaker: Studio.
8. Genymotion.

1. Android Studio

There's no talking about android app development without the Android Studio. It's the most basic tool for Android developers. Created by Google in 2013, it has pretty much become the standard software for Android Developers. It's a great tool because it has the support of Google as well as a large community of developers.

2. Android Debug Bridge (ADB)

Android Debug Bridge is included in Android Studio and it's basically a line of communication between Android devices and other computers that developers use for QA and testing purposes. Android Developers can connect their Android device to their computer and make necessary changes to both devices at the same time.

3. Android Virtual Device (AVD) Manager

Another great feature of Android Studio is the AVD. This is an emulator that will run your Android app on your computer so that you have a better inside into what your code looks like. It's great for actually seeing the work you've done and making any adjustments as needed.

4. Eclipse

Next up on the list we have Eclipse. Before Android Studio came around, Eclipse was the main tool for Android development. As of right now, Google doesn't support this software but some developers still use it to build Android apps as well as other apps. Eclipse is still a pretty useful tool, especially for developing cross-platform applications, and it supports a variety of programming languages.

Types of Android Application Development Kits

1. **Android Software Development Kit (SDK)**
2. **Android Native Development Kit (NDK)**

Its libraries are written in C, C++ & other compiler languages & it can be installed by developers using the Android Native Development Kit. Android NDK classes can be called from Java (to be precise).

3. **Android Open Accessory Development Kit (OAK)**
   The Android Open Accessory Development Kit has been provided to developers from Android 3.1 versions and above which allows USB hardware to interact with
   Android-powered phones when it is primarily in the "Accessory Mode". The so-called as the connected accessory acts as a USB host & the device acts as a USB device in this scenario. OADK allows to detect Android devices that are in "Accessory Mode" only (& not in any other mode).

Developer Tools

The android developer tools let you create interactive and powerful application for android platform. The tools can be generally categorized into two types.

1. SDK tools

   SDK tools are generally platform independent and are required no matter which android platform you are working on. When you install the Android SDK into your system, these tools get automatically installed.
   - Provides the Java framework classes
   - Compiles to java bytecode
   - Class framework is updated with every OS release

   1. android
   This tool lets you manage AVDs, projects, and the installed components of the SDK

   2. ddms
   This tool lets you debug Android applications

   3. Draw 9-Patch
   This tool allows you to easily create a NinePatch graphic using a WYSIWYG editor

   4. emulator
   This tools let you test your applications without using a physical device

   5. mksdcard
   Helps you create a disk image (external sdcard storage) that you can use with the emulator

6. proguard

Shrinks, optimizes, and obfuscates your code by removing unused code

7. sqlite3

Lets you access the SQLite data files created and used by Android applications

8. traceview

Provides a graphical viewer for execution logs saved by your application

9. Adb

Android Debug Bridge (adb) is a versatile command line tool that lets you communicate with an emulator instance or connected Android-powered device.

2. **Platform tools**

The platform tools are customized to support the features of the latest android platform.

- runs and debugs apps from your dev machine

The platform tools are typically updated every time you install a new SDK platform. Each update of the platform tools is backward compatible with older platforms.

Some of the platform tools are listd below −

1) Android Debug bridge (ADB)

2) Android Interface definition language (AIDL)

3) aapt, dexdump , and dex e.t.c

The Android SDK Platform Tools are a separately downloadable subset of the full SDK, consisting of command-line tools such as adb and fastboot.

CHAPTER – 5

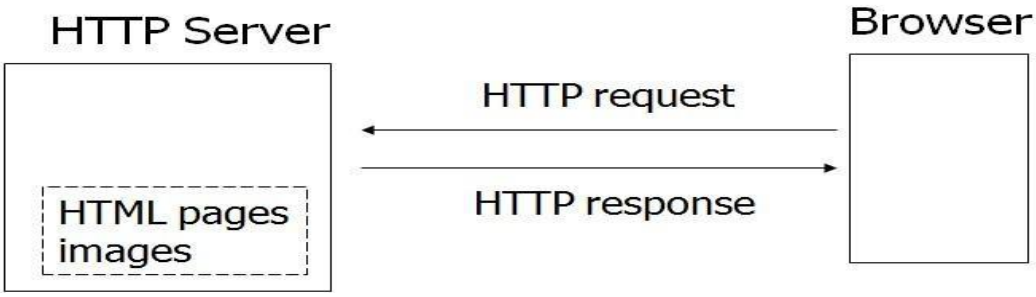
Introduction To Web Applications

A web application is a client-side and server-side software application in which the client runs or request in a web browser. Common web applications include email, online retail sales, online auctions, wikis, instant messaging services and more.

A web application (or web app) is application software that runs on a web server, unlike computer-based software programs that are run locally on the operating system (OS) of the

device. Web applications are accessed by the user through a web browser with an active network connection.

These applications are programmed using a client–server modeled structure—the user ("client") is provided services through an off-site server that is hosted by a third-party.



Its frontend is usually created using languages like HTML, CSS, Javascript, which are supported by major browsers.

History of web Application:

A web application (or web app) is application software that runs on a web server, unlike computer-based software programs that are run locally on the operating system (OS) of the device. Web applications are accessed by the user through a web browser with an active network connection. These applications are programmed using a client–server modeled structure—the user ("client") is provided services through an off-site server that is hosted by a third-party. Examples of commonly-used web applications include: web-mail, online retail sales, online banking, and online auctions.

The first workable prototype of the Internet came in the late 1960s with the creation of ARPANET, or the Advanced Research Projects Agency Network. Originally funded by the U.S. Department of Defense, ARPANET used packet switching to allow multiple computers to communicate on a single network.

The year 1995 is a crucial year in the era of the Internet. Netscape Communications presented JavaScript, a client-side scripting language that enables programmers to improve the user interface with the dynamic elements. JavaScript made the Internet faster and more productive because the data was no longer sent to the server to generate the whole web page.
In 1996, Macromedia Flash was introduced. It was also a revolutionary innovation that made the Web 'brighter' and interactive. This vector animation player enabled programmers to enrich web pages with animation. This multimedia software platform works with animation, different types of browser games, vector graphics and Internet and mobile applications.

In 1999, the concept of web application appeared in Java language. Later on, in 2005, Ajax was introduced by Jesse James Garrett in his article "Ajax: A New Approach to Web Application". This complex of web development techniques enabled programmer to compose asynchronous web apps.

Platforms used for web applications

1. **WordPress**
   WordPress is a content management system (CMS) that allows you to host and build websites. WordPress contains plugin architecture and a template system, so you can customize any website to fit your business, blog, portfolio, or online store.
2. **PHP**
   PHP files can contain text, HTML, CSS, JavaScript, and PHP code. PHP code is executed on the server, and the result is returned to the browser as plain HTML. PHP files have extension ".php".
   - PHP can generate dynamic page content
   - PHP can create, open, read, write, delete, and close files on the server
   - PHP can collect form data
   - PHP can send and receive cookies
   - PHP can add, delete, modify data in your database
   - PHP can be used to control user-access
   - PHP can encrypt data

3. **AngularJS. AngularJS Framework By Google It was first released in 2009, and completely rewritten in 2016.**
   AngularJS Framework By Google It was first released in 2009, and completely rewritten in 2016. Angular.JS is a powerful JavaScript-based development framework. It is an open source framework. This Framework is used by companies like Google, Paypal, Netflix, Upwork, Microsoft and other Giants.
   AngularJS can be Considered the best Front-end framework for a web application. Angular JS is great for Video Streaming Apps, Travel Apps, Weather Apps, eCommerce, Mobile Commerce, Social Apps, User-Review Applications. It is used to create dynamic web applications.
4. **ASP.NET. Asp.net was Established in the year 2002.**
   Asp.net was Established in the year 2002. It is the most Popular Framework built by Microsoft. Asp.net is Perfect Web app development platform for building desktop and mobile applications. Asp.Net is very useful for creating Dynamic and scalable web applications.
5. **Ruby on Rails. ROR framework written in the Ruby language.**
   ROR framework written in the Ruby language. Ruby on Rails is an open-source web application framework. Rails help you build websites. It comes with a large pool of reference library and great tools to create web apps. Ruby is growing very quickly. A few big names that used Ruby on Rails Basecamp, GitHub, Shopify, Airbnb,
6. **Node**

Node.js is the best when you build a highly efficient website. It is a very powerful JavaScript-based platform. You can develop web applications like video streaming sites, single page applications, and many other applications. As technology is changing developers are moving to adopt convenient frameworks for their web development needs. **7. React**

React, Known as ReactJS an open source Javascript library, that is used to create websites. React.JS is maintained by Facebook. Reactjs helps to build SEO friendly and highperformance Web applications. React is definitely worthy especially if you are

working on a single page application. you can make web apps user-friendly, responsive and super fast with ReactJs.

Components of web applications

All web-based database applications have three primary components: A web browser (or client), a web application server, and a database server.

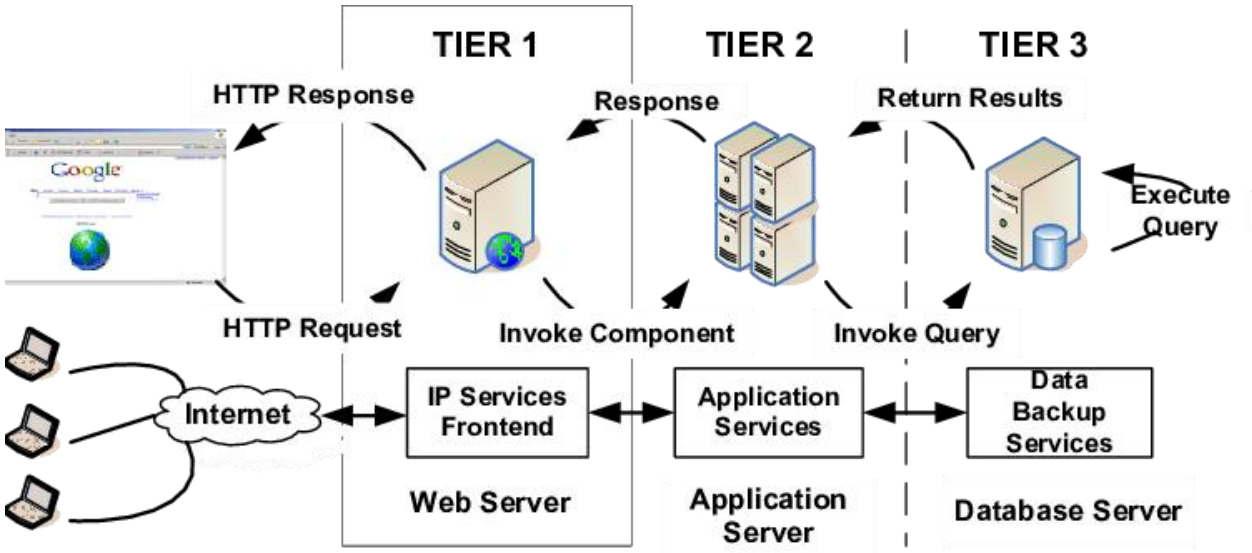
Working

They are basically used in a web-based application that has 3 tier architecture. The position at which the application server fits in is described below:

Tier 1 – This is a GUI interface that resides at the client end and is usually a thin client (e.g. browser)

Tier 2 – This is called the middle tier, which consists of the Application Server.

Tier 3 – This is the 3rd tier which is backend servers. E.g., a Database Server.



Web-based database applications rely on a database server, which provides the data for the application. The database server sometimes also provides business logic in the form of stored procedures. Stored procedures can offer significant performance advantages, especially in a multi-tiered architecture.

Three primary components are:

1) Web Browser

A web browser (also referred to as an Internet browser or simply a browser) is application software for accessing the World Wide Web or a local website. When a user requests a web page from a particular website, the web browser retrieves the necessary content from a **web server** and then displays the page on the user's device. **Working**

The purpose of a web browser is to fetch content from the Web or from a local storage device and display it on a user's device.

This process begins when the user inputs a Uniform Resource Locator (URL), such as https://bhanudaya.edu.np/, into the browser. Virtually all URLs on the Web start with either http: or https: which means the browser will retrieve them with the Hypertext Transfer Protocol (HTTP). In the case of secure mode (HTTPS), the communication

between the browser and the web server is encrypted for the purposes of security and privacy. Once a web page has been retrieved, the browser's rendering engine displays it on the user's device. This includes image and video formats supported by the browser.

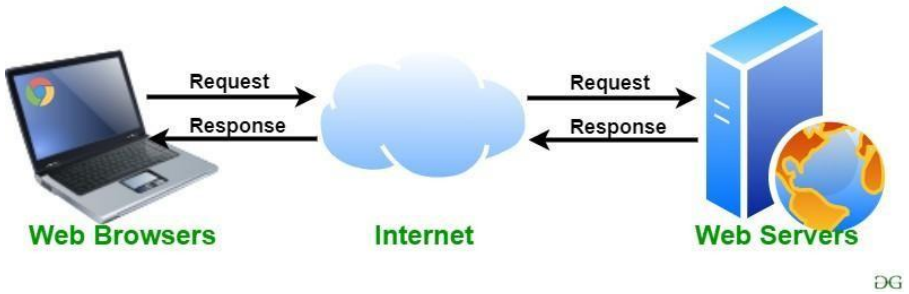
Difference between Web Browser and Web Server

Web Browser	Web Server
Web Browser is an Application program that displays a World wide web document. It usually uses the internet service to access the document.	Web server is a program or the computer that provide services to other programs called client.
The Web browser requests the server for the web documents and services.	The Web server accepts, approve and respond to the request made by the web browser for a web document or services.
The web browser act as an interface between the server and the client and displays a web document to the client.	The web server is a software or a system which maintain the web applications, generate response and accept clients data.

The web browser sends an HTTP request and gets an HTTP response.	The web server gets HTTP requests and send HTTP responses.

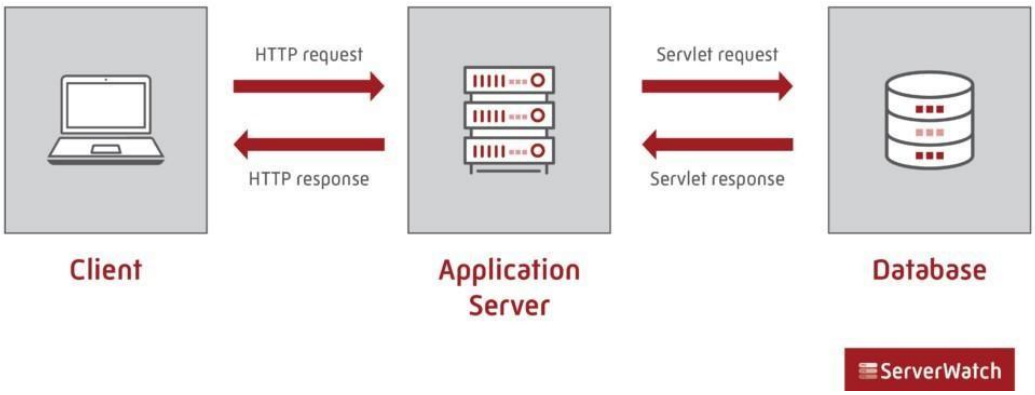
Web Browser	Web Server
Doesn't exist any processing model for the web browser.	There exist three types of processing models for web server i.e Process-based, Thread based and Hybrid.
Web browser stores the cookies for different websites.	Web servers provide an area to store and organize the pages of the website.
The web browser is installed on the client computer.	The web server can be a remote machine placed at the other side of your network or even on the other end of the globe, or it is your very own personal computer at home.

Web Server



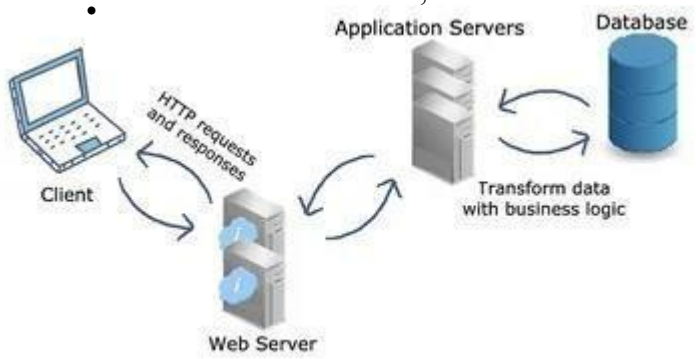
2) Application server

An application server is a modern form of platform middleware. It is system software that resides between the operating system (OS) on one side, the external resources (such as a database management system [DBMS], communications and Internet services) on another side and the users' applications on the third side.



Application servers consist of:

- web server connectors,



An application server is a server that hosts applications.
Application server frameworks are software frameworks for building application servers. An application server framework provides both facilities to create web applications and a server environment to run them.

- computer programming languages,
- runtime libraries,
- database connectors, and
- the administration code needed to deploy, configure, manage, and connect these components on a web host.

3) Database Server

Database servers are used to store and manage databases that are stored on the server and to provide data access for authorized users. This type of server keeps the data in a central location that can be regularly backed up. It also allows users and applications to centrally access the data across the network. A large number of the databases used in your organization can be kept on one server or a group of servers that are specifically configured to protect data and service client requests.

Users access a database server either through a "front end" running on the user's computer – which displays requested data – or through the "back end", which runs on the server and handles tasks such as data analysis and storage.

